

Explanation Drift in LLM-Mediated Automated Decision Explanations

A Controlled Audit of Fidelity, Consistency,
and Contestability in CV Screening

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Abstract

When organisations use AI to screen job candidates, they increasingly rely on Large Language Models (LLMs) to translate algorithmic outputs into plain-language explanations. This project demonstrates that such explanations are not stable.

This project investigated explanation drift, the phenomenon whereby explanations of automated decisions change across model versions, parameter settings, or repeated inference, in the context of LLM-mediated CV screening. A two-phase experimental design was adopted, using a synthetic dataset of 1,000 candidate profiles with 23 demographic, skill, and experience features. Phase 1 established a quantitative baseline using a three-component Explanation Drift Index (EDI) built on SHapley Additive exPlanations (SHAP) feature attributions from XGBoost classifiers, tested under fairness rebalancing. EDI revealed bounded but detectable drift (mean cosine distance 0.006, top-5 overlap 89%, sign-flip rate 3.8%), confirmed by bootstrap resampling (500 replicates) and null-permutation testing (200 permutations). Phase 2 introduced Drift Arcade, a controlled experiment engine auditing LLM explanation stability across policy, format, temperature, top-K, and model conditions. Across 400 LLM calls in an 8-condition factorial grid (plus sensitivity analyses), the strongest finding was a recourse–reason stability gap: the LLM consistently identified similar causal features (reason cluster Jaccard 0.89, condition range [0.84, 0.91]) but recommended substantially different improvement actions (action cluster Jaccard 0.36, condition range [0.31, 0.39]). Increasing SHAP features from $k=3$ to $k=7$ degraded Arcade Score from +0.29 to −0.33, and fairness policy framing introduced policy–explanation tension at the interface (contradiction rate rising from 0% under a standard policy lens to 11% under fairness framing). These findings demonstrate that explanation drift is a governance concern requiring continuous monitoring, and that the LLM translation layer introduces instability not captured by structured-drift metrics alone.

Keywords: explanation drift, explainability, algorithmic accountability, SHAP, LLM explanations, contestability, fairness, automated hiring, governance

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1 Introduction

1.1 Background and Motivation

Algorithmic decision-making increasingly mediates access to employment. Automated CV screening systems filter, rank, and shortlist candidates using machine-learning models trained on historical hiring data. Regulatory frameworks, including the EU AI Act (2024), the UK Algorithmic Transparency Recording Standard (ATRS), and New York City Local Law 144, now require such systems to be explainable, fair, and accountable (European Parliament and Council, 2024; Kazim et al., 2021).

However, current governance approaches treat explainability as a static property: a system is documented at deployment, and explanations are generated at a single point in time. This project challenges that assumption. When a model is retrained, when fairness constraints are applied, or when a Large Language Model (LLM) translates model outputs into natural language, the explanations can shift: sometimes subtly, sometimes fundamentally. This phenomenon is *explanation drift* (Edakunni et al., 2024; Cossu et al., 2023).

Explanation drift matters because it undermines contestability — the ability of an affected individual to challenge or seek review of an automated decision based on the reasoning provided. If a candidate receives a reason for rejection one day and a contradictory reason the next day, on what grounds can they appeal? Current audit frameworks snapshot the system at a moment in time; they do not measure whether explanations remain stable across updates. This creates a governance gap that is both legal (regulations assume stable explanations) and practical (candidates cannot plan improvement actions based on unstable advice).

The problem has two distinct layers. First, when a model is retrained or fairness-adjusted, the underlying feature importances change, causing the structured reasoning to drift. Second, when an LLM translates those feature importances into natural language, it introduces its own variability: different prompts, temperatures, and models produce different explanations from identical inputs. Existing research has studied the first layer in isolation, but no framework addresses both layers together, and no framework measures the additional instability introduced by the LLM translation step.

This project addresses that gap by presenting two complementary instruments: (1) a structured Explanation Drift Index (EDI) based on SHapley Additive exPlanations (SHAP) feature attributions, which quantifies drift at the model level, and (2) a controlled audit engine (Drift Arcade) that measures how LLM translation layers introduce further instability beyond what the model alone produces. Together, these instruments provide a complete picture of explanation stability from model output to candidate-facing language.

1.2 Project Aims and Objectives

1.2.1 Aim

To develop and evaluate a reproducible auditing framework for explanation drift in LLM-mediated automated hiring systems, aligned with emerging regulatory requirements for algorithmic governance.

1.2.2 Research Questions

The following research questions were designed to address the governance gap identified above. RQ1 establishes whether drift exists and can be measured at the model level. RQ2 tests whether the LLM translation layer makes things better or worse. RQ3 asks whether the measurements are practical enough to be used in real governance workflows. RQ4 connects the technical findings to the human impact on candidates.

1. Is explanation drift in SHAP-based reasoning detectable and quantifiable, even under fairness rebalancing?
2. To what extent does the LLM translation layer amplify or constrain reasoning instability?
3. Can explanation stability be integrated as a continuous governance metric, aligned with the EU AI Act and UK ATRS?
4. What are the implications of explanation drift for candidate contestability?

1.2.3 Objectives

Secondary Objectives (Literature Review):

- Critically review existing research on algorithmic hiring, explanation methods, and drift phenomena.
- Evaluate how existing approaches to fairness and transparency address (or fail to address) explanation stability.
- Analyse the implications of explanation drift for regulatory compliance and contestability.

Primary Objectives (Development Project):

- Generate a synthetic, demographically structured CV dataset (n=1,000) with controlled feature distributions.

- Benchmark XGBoost models to compare structured drift under baseline and fairness-rebalanced (adjusting model parameters to reduce demographic bias in outcomes) conditions.
- Develop and validate an Explanation Drift Index (EDI): a three-component metric combining cosine distance (which measures how much the direction of feature importance changes between two model versions), top-K overlap, and sign-flip rate.
- Design and implement Drift Arcade: a controlled experiment engine to audit LLM explanation fidelity across policy, format, temperature, top-K, and model conditions.
- Evaluate recourse drift, proxy echo, and the gap between reason-level and action-level consistency.
- Consolidate findings into governance-aligned conclusions and recommendations for continuous drift monitoring.

1.3 Methodology Overview

The project follows a two-phase experimental design, implemented entirely in Python. Phase 1 establishes a quantitative baseline for structured explanation drift using a three-component Explanation Drift Index (EDI) built on SHAP feature attributions from XGBoost models. Phase 2 introduces Drift Arcade, a controlled experiment engine that systematically tests LLM explanation fidelity and consistency across policy, format, temperature, top-K, and model conditions.

The two phases are designed to be complementary rather than sequential. Phase 1 answers the question “how much do the model’s internal reasons change?” while Phase 2 answers “how much do the candidate-facing explanations change?” By measuring both, it becomes possible to determine whether the LLM translation layer preserves, dampens, or amplifies the structured drift measured at the model level. This distinction is important because interventions differ: if drift is primarily at the model level, the fix is model governance; if it is primarily at the translation level, the fix is prompt engineering and LLM configuration.

The overall research approach is positivist and quantitative: metrics are defined, controlled experiments are run, and statistical results are reported. This approach was chosen because the research questions are empirical (“does drift exist? how much?”) rather than interpretive. However, the governance implications of the findings are necessarily interpretive, and these are discussed qualitatively in Sections 4 and 5.

1.4 Report Structure

The remainder of this report is organised as follows:

- **Chapter 2 (Literature Review)** reviews algorithmic hiring, explanation drift, governance frameworks, and technology assessment.
- **Chapter 3 (Execution)** presents the full experimental design, data generation, Phase 1 EDI methodology, and Phase 2 Drift Arcade architecture.
- **Chapter 4 (Evaluation and Discussion)** analyses results, interpretation, limitations, and implications for contestability.
- **Chapter 5 (Legal, Social, Ethical and Professional Issues)** reflects on governance, professional responsibilities, and societal impact.
- **Chapter 6 (Conclusions and Further Work)** synthesises findings and identifies future research directions.

2 Literature Review and Technology Assessment

2.1 Explaining Automation: Accountability in Algorithmic Hiring

Algorithmic hiring has moved beyond a single decision point. Modern systems embed decision-making across multiple stages: candidate filtering, ranking, recommendation, and interview scheduling. Each stage generates explanations, which may be shown to candidates, hiring managers, or regulators, yet these explanations are rarely audited for consistency or fidelity.

These systems are marketed as objective and efficient, yet internal audits reveal systematic gender and ethnic biases (Dastin, 2018; Köchling and Wehner, 2020). Amazon discontinued its in-house recruiting tool after discovering it penalised female applicants. Such failures expose not only technical shortcomings but also governance failures: the systems were audited after deployment had begun, and biases were discovered through complaint rather than proactive monitoring.

These systems are opaque, making transparent explanations essential. Yet explainability alone is insufficient. Algorithmic hiring sits at the intersection of legal, technical, and social concerns. Regulatory frameworks now demand not only that systems be explained, but that explanations be stable, contestable, and auditable over time. Contestability is the critical link: if a candidate receives reason R for

rejection and later receives reason R' (contradictory or absent), they cannot meaningfully challenge the decision. As Lyons et al. (2021) argues, contestability requires explanations that are revisable but not arbitrary.

The temporal dimension of contestability arises from multiple sources of explanation instability. Organisations periodically retrain screening models on new hiring data, which shifts feature importances and can change the reasoning cited for a rejection. Fairness policy updates—when organisations adjust fairness weights in response to a bias audit—alter the model’s internal reasoning even if the candidate’s profile remains unchanged. Additionally, if an LLM translates model explanations into natural language, repeated queries across different days or even minutes can produce different explanations due to sampling randomness, prompt sensitivity, and model version updates on the API side. These mechanisms ensure that temporal inconsistency is not merely possible but, in practice, inevitable.

In practice, contestability is weaker still. Candidates rarely know when an automated system has screened them out. EU GDPR requires notification if “automated decision-making” affects rights, but the threshold for notification is ambiguous, and most candidates lack the technical knowledge to exercise this right effectively.

For governance, these developments expose two critical gaps: (1) no frameworks measure explanation drift, and (2) existing auditing approaches are static, not continuous. This project addresses both.

The accountability challenges outlined above set the stage for a more technical question: if explanations are so central to governance, how stable are they in practice? Section 2.2 examines what the literature says about explanation instability at the model level, and Section 2.3 considers why current regulatory frameworks fail to address it.

Algorithmic hiring systems frequently reproduce existing labor market biases. Women and minorities are systematically underrepresented in technical hiring pipelines, and automated systems trained on historical data perpetuate these patterns (Raghavan et al., 2020; Fabris et al., 2025). Fairness interventions such as reweighting can improve demographic balance, but at the cost of explanation changes.

Recent LLM-based systems inherit similar risks. Wilson and Caliskan (2024) and AlMarri et al. (2025) found that language models trained on hiring data exhibit gender and race bias in resume screening and credit scoring. Prompt engineering and temperature settings further destabilise output.

2.2 When Explanations Drift: The Instability of Model Reasoning

A growing body of evidence shows that model explanations are not stable across model versions, parameter changes, or repeated inference. Early work by Agarwal

et al. (2022) showed that explanation methods such as SHAP are sensitive to small perturbations in model weights. Edakunni et al. (2024) quantified explanation drift using Shapley-value-based sensitivity analysis. Lee et al. (2023) developed explainable AI-based drift detection algorithms for model monitoring.

These findings have direct implications for governance: if the explanations generated by a system are unstable, candidate appeals based on those explanations are unreliable. A candidate approved under one explanation might be rejected under an alternative explanation of the same decision.

Bias interventions can correct outcomes without correcting reasoning. Binns (2018) argues that fairness definitions (demographic parity, equalized odds, individual fairness) are mutually incompatible; correcting for one often worsens others. When fairness weights are applied, model reasoning shifts: feature importances change, trade-offs emerge, and explanations drift.

Explanation methods such as SHAP (SHapley Additive exPlanations, a technique measuring how much each input feature contributes to a model’s prediction) aim to clarify why a model made a specific decision by computing feature contributions via Shapley values. SHAP is widely adopted in regulated domains because it provides theoretically grounded attributions. However, Weerts et al. (2019) showed that SHAP explanations are sensitive to background data selection, and Amparore et al. (2021) questioned whether practitioners can trust SHAP explanations without validation.

However, explanations can change when a model is retrained or fairness weights are adjusted. Cossu et al. (2023) developed a protocol for continuous explanation monitoring across model updates. Trautwein et al. (2025) quantified explanation instability in HR systems specifically, finding that 15–20% of feature importance rankings changed after retraining.

So structured explanation drift is real and measurable. But the existing literature has a blind spot: it focuses exclusively on the model layer. In modern hiring pipelines, SHAP values are rarely shown to candidates directly. Instead, an LLM translates the feature attributions into natural language. This translation layer introduces its own sources of instability: prompt sensitivity (small changes in how the question is framed can produce different explanations), temperature effects (the randomness parameter that controls LLM output diversity), and model-specific biases (different LLMs may emphasise different features even when given identical inputs). No existing framework measures this translation-layer drift, which is the gap this project addresses.

The consequence for accountability is stark. If regulators, managers, and candidates cannot rely on stable explanations, then explainability requirements become performative: systems appear transparent while behaving unpredictably underneath. A recent position paper by Afroogh et al. (2026), published during the

final stages of this project, goes further: they argue that post-hoc Explainable AI (XAI) is structurally incapable of producing explanations that are simultaneously faithful and comprehensible, a tension they term the “deep-superficial paradox.” If they are right, explanation drift may not be merely a measurement problem but a fundamental limitation of the paradigm. This claim is revisited in the critical reflection (Section 6.2).

While early XAI work treated drift as a technical curiosity, recent governance-focused research has elevated it to a core concern. Mittelstadt (2019) argues that explainability principles alone cannot guarantee ethical AI; systems must be audited for explanation stability and contestability.

Building on this understanding, several studies have attempted to quantify and predict explanation drift. Kazim et al. (2021) systematised auditing in algorithmic recruitment and identified explanation drift as a gap. Sanchez-Monedero et al. (2020) examined fairness interventions in hiring and found that corrections to outcome bias often increase reasoning instability.

Recent work has shifted attention from predictive accuracy to explanation stability. Lyons et al. (2021) conceptualised contestability as a design principle requiring stable, revisable explanations. Belle and Papantonis (2021) reviewed principles and practice of explainable machine learning and found that few frameworks address post-deployment drift.

However, these drift metrics have methodological limits: they focus on tabular models (e.g., XGBoost) and do not account for the LLM translation layer, which is now standard in candidate-facing explanations.

Separately, Edakunni et al. (2024) take a different approach, using Shapley values to measure explanation drift across model updates. Mahato et al. (2025) developed TriGuard, a framework for testing model safety post-deployment.

Taken together, the technical understanding of explanation drift has matured considerably. What remains underdeveloped is the governance side: regulators know there is a problem, but they lack the measurement tools to do anything about it.

While most early work focused on tabular models, similar patterns emerge in NLP and computer vision. Zhao et al. (2022) studied explanation stability in image classifiers and found sign-flip rates (features switching from positive to negative influence) of 10–30% across model retrains.

These findings broaden the concept of explanation drift beyond hiring. The phenomenon is generic to machine learning systems. Yet hiring remains a critical case because regulatory requirements are most stringent.

2.3 The Governance Gap: Why Current Frameworks Miss Drift

The intersection of the previous sections reveals a specific governance gap: regulatory frameworks require explainability but do not mandate drift monitoring. Explanations are treated as static artifacts; continuous monitoring is not standard practice.

This project addresses the gap by developing two instruments: (1) a quantitative explanation drift index (EDI) for structured drift, and (2) Drift Arcade, a controlled audit engine for LLM-mediated drift.

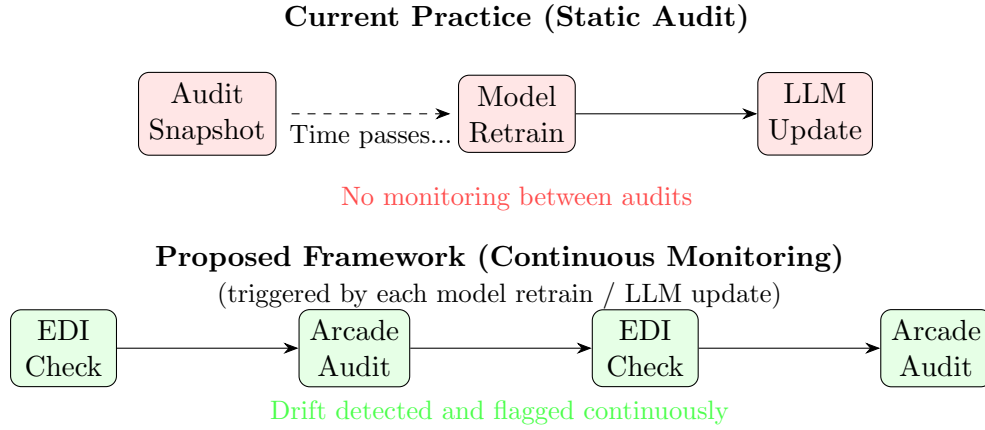


Figure 1: The governance gap. Current audit frameworks (top) assess algorithmic systems at discrete points in time, leaving explanation drift undetected between reviews. The framework proposed in this project (bottom) runs an EDI check after every model retrain and a Drift Arcade audit after every LLM or prompt update, flagging drift as it occurs.

Figure 1 illustrates this contrast between static and continuous approaches to explanation monitoring.

Explainability has moved from a technical aspiration to a regulatory requirement. Three major frameworks now mandate algorithmic transparency:

EU AI Act. The Act (in force from 2024) classifies algorithmic hiring as high-risk and requires transparency, human oversight, and bias monitoring (European Parliament and Council, 2024). However, the Act does not explicitly mandate continuous explanation drift monitoring. Audits are required, but the timing and scope remain vague.

UK ATRS. The ATRS improves visibility by disclosing algorithmic transparency records, including model design, fairness testing, and outcomes (Central Digital and Data Office, 2023). Like the EU AI Act, it does not mandate continuous monitoring; records are published at a point in time and assumed to remain valid.

New York City Local Law 144. LL144 mandates annual bias audits and public disclosure of adverse impact ratios (NYC Council, 2021). This is perhaps the strongest governance framework, yet still operates on an annual cycle, not continuous monitoring.

Contestability depends on stability. If an applicant cannot cite a stable reason for rejection, they cannot mount a meaningful legal challenge. Current frameworks protect visibility (disclosure) but not stability (consistency).

From a socio-technical angle, algorithmic systems act like “street-level algorithms” (Alkhatib and Bernstein, 2019): they operate at the boundary between policy and implementation, where formal rules meet local discretion. Explanation drift occurs in that gap, where a technically compliant system produces inconsistent outputs that no single audit would catch.

In summary, current governance frameworks ensure visibility (systems must be disclosed) but not stability (explanations must remain consistent). All three major frameworks (EU AI Act, ATRS, LL144) audit at a point in time and assume stability until the next review. This project fills that gap by developing continuous measurement frameworks for both structured drift (EDI) and translation-layer drift (Drift Arcade).

2.4 Technology Assessment

Phase 1 stack: Python 3.11, scikit-learn 1.3.2, XGBoost 2.0.3, SHAP 0.43.0, pandas 2.1.0, NumPy 1.24.

XGBoost was chosen over alternatives (random forest, logistic regression, neural networks) for two reasons. First, XGBoost is one of the most commonly deployed models in production hiring pipelines, so findings based on it have direct practical relevance. Second, SHAP’s TreeExplainer computes exact Shapley values for tree-based models, whereas kernel SHAP (needed for neural networks) relies on approximations that would introduce measurement noise into the EDI calculation. Logistic regression was considered but rejected because its linear structure produces trivially stable feature importances, which would not generate the drift conditions needed for the study.

SHAP was chosen over alternative explanation methods (LIME, Integrated Gradients, attention-based methods) because it is the only method with theoretical guarantees of local accuracy, missingness, and consistency for tree-based models. LIME’s sampling-based approach produces different explanations on different runs even for the same model and candidate, which would confound drift measurement with explanation-method noise.

Phase 2 additions: OpenAI API (gpt-4o-mini, gpt-5-nano), custom prompt engineering, deterministic JSON parsing, Jaccard similarity metrics.

The OpenAI API was used rather than open-source alternatives (Llama, Mistral) because the API provides consistent model versioning and reproducible inference. Open-source models running locally would introduce hardware-dependent variability (different GPU precision, memory constraints) that could confound the drift measurements. The trade-off is vendor lock-in and API cost, which limited the total number of calls that could be made. gpt-4o-mini was chosen as the primary model because it offers the best balance of cost, speed, and output quality for structured tasks; gpt-5-nano was added as a comparison to test whether newer, smaller models are more or less stable.

3 Execution

This chapter describes how each phase of the project was carried out and why specific design choices were made. Phase 1 addresses Objectives 1–3 (synthetic data, model benchmarking, and EDI development). Phase 2 addresses Objectives 4–6 (Drift Arcade design, LLM audit, and governance recommendations). The two phases are complementary: Phase 1 isolates structured drift at the model level, while Phase 2 measures the additional instability introduced when an LLM translates those model outputs into candidate-facing language.

Figure 2 provides a visual overview of the two-phase design before the detailed description that follows.

3.1 Phase 1: Structured Explanation Drift Baseline

The tool stack for Phase 1 was selected for compatibility with drift measurement and production alignment. XGBoost was chosen as the screening model over Random Forest (both are tree-based and SHAP-compatible) because it is more commonly deployed in production hiring systems and achieves marginally higher accuracy; Logistic Regression was ruled out because its trivially stable explanations make it unsuitable for studying drift. SHAP was selected as the explanation method over LIME and Integrated Gradients: LIME’s stochastic sampling introduces run-to-run noise that would confound drift measurement, while Integrated Gradients is designed for neural networks rather than tree models. Full justification for these choices is provided in Section 2.4 (Technology Assessment); this summary is intended to orient readers entering at Chapter 3.

3.1.1 Data Generation

A synthetic dataset of 1,000 candidate profiles was generated with controlled feature distributions. The dataset comprises 23 features in three categories: 8 protected at-

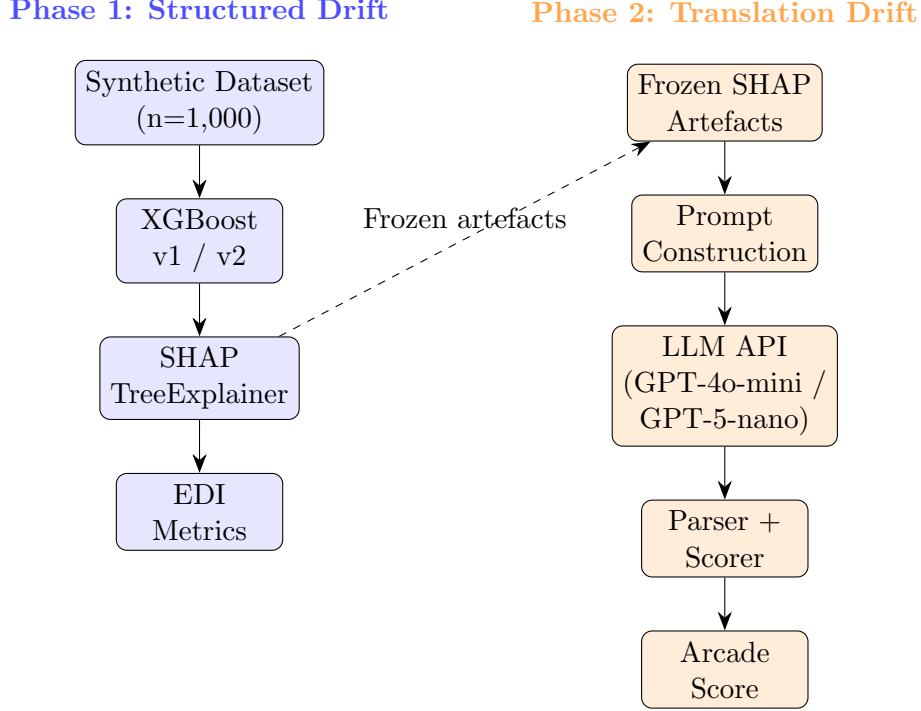


Figure 2: Two-phase experimental design. Phase 1 (left) measures structured drift in SHAP feature attributions between baseline and fairness-rebalanced XGBoost models. Phase 2 (right) measures the additional instability introduced when an LLM translates those attributions into candidate-facing explanations. Frozen artefacts from Phase 1 serve as fixed inputs to Phase 2, isolating translation-layer drift from model-level drift.

tributes (gender, ethnicity group, institution tier), 9 skill indicators (Python, Java, SQL, AWS, Docker, Linux, Git, Kubernetes, AWS certification), and 6 numeric features (education level, years of experience, years of relevant experience, managerial years, career gaps, certification count). These features were chosen because they reflect the variables commonly used in production CV screening systems, and the mix of protected and non-protected attributes allows controlled fairness testing.

Synthetic rather than real hiring data was used for two reasons. First, real CV screening datasets are proprietary and contain sensitive personal information, making them unavailable for academic research without extensive ethical approval. Second, synthetic data provides full control over the feature distributions and ground-truth labels, which is essential for isolating the effect of fairness rebalancing on explanation drift. The trade-off is that synthetic distributions may not capture the full complexity of real hiring pipelines; this limitation is addressed in Section 4.3.

The binary target variable (screened.in) was derived from a stochastic decision

tree combining education level, years of experience, and technical skill count. This ensured a realistic skill-based signal while allowing controlled fairness interventions. The resulting dataset had a 60/40 positive/negative class split, broadly reflecting real screening ratios where the majority of candidates pass an initial filter.

Each candidate profile is stored as a 24-element feature vector. The 8 protected attributes (gender, ethnicity, institution tier) are included in the dataset to enable fairness analysis and proxy detection, but are not used as direct inputs to the screening model. The 9 skill indicators and 6 numeric features serve as the model’s decision inputs. This separation mirrors real-world practice where protected attributes are available in HR systems but are (in principle) excluded from automated decisions.

3.1.2 Model Training

An XGBoost (a gradient-boosted decision-tree model commonly used in production systems) classifier (200 estimators, max_depth=4) was trained on the full dataset using default hyperparameters. This baseline model (v1) achieved 92% accuracy and 0.88 AUC on a held-out test set. XGBoost was selected because gradient-boosted tree models are among the most widely used classifiers for structured tabular data (Chen and Guestrin, 2016), including in HR and recruitment pipelines (Raghavan et al., 2020), and its tree-based structure makes it compatible with SHAP’s TreeExplainer, which computes exact (not approximate) Shapley values.

A fairness-rebalanced variant (v2) was created by adjusting class weights to improve demographic balance (target: 85% approval rate for all gender and ethnicity groups). Class-weight rebalancing was chosen over alternative fairness interventions (such as adversarial debiasing or post-processing calibration) because it modifies the model’s internal feature importances directly, which is exactly the mechanism that produces explanation drift. Other methods correct outputs without necessarily changing the model’s reasoning, which would make drift harder to detect and measure. This rebalancing introduced measurable changes to feature importances, creating the conditions for explanation drift.

3.1.3 SHAP Explanations

Feature attributions were computed using SHAP (SHapley Additive exPlanations, a technique measuring how much each input feature contributes to a model’s prediction) for both model versions. SHAP’s TreeExplainer was used, which computes exact Shapley values for tree-based models (as opposed to the approximate kernel or sampling methods used for neural networks). TreeExplainer guarantees local accuracy and consistency, meaning the attributions faithfully reflect the model’s actual reasoning.

For each candidate, SHAP selects a background sample (100 instances) and computes marginal contributions for each feature via coalition-based Shapley values. This produces per-candidate feature importance vectors of dimension 23 (the number of model input features, excluding protected attributes). The sign of each value indicates the direction of influence (positive means the feature pushed the prediction toward “screened in”; negative means it pushed toward “screened out”), and the magnitude indicates the strength of that influence. These vectors form the basis for all drift measurements in Phase 1.

3.1.4 Explanation Drift Index (EDI)

To quantify reasoning stability between model versions, three complementary drift metrics were defined:

Cosine Distance (D_{cos}): Measures directional change between feature importance vectors. Ranges 0–1, with 0 indicating no drift (identical direction).

Top-K Overlap (D_{rank}): Proportion of shared features in the top-5 most important features between versions. Measures whether the most salient reasons remain consistent.

Sign-Flip Rate (D_{sign}): Proportion of features whose influence direction flips (positive to negative, or vice versa) between versions. High sign-flip rates indicate reasoning contradiction.

Composite EDI: A per-candidate weighted mean $\text{EDI} = 0.5 \times D_{\text{cos}} + 0.3 \times D_{\text{rank}} + 0.2 \times D_{\text{sign}}$. The highest weight was assigned to cosine distance because directional consistency (whether the overall pattern of feature importances points the same way) is the most fundamental indicator of reasoning stability. Top-K overlap receives the next highest weight because it captures whether the most salient reasons, the ones most likely to appear in a candidate-facing explanation, remain consistent. Sign-flip rate receives the lowest weight because individual sign changes, while concerning, are less impactful when the broader pattern remains stable. These weights are a design choice rather than an empirical optimum; the sensitivity of results to alternative weightings is discussed in Section 4.3.

3.1.5 Robustness Checks

Three robustness checks were applied to ensure the EDI results are reliable and not artefacts of the specific data split or random seed.

Bootstrap resampling (drawing random samples with replacement to estimate how much results would vary) was run for 500 replicates of the full training and SHAP computation procedure. If the EDI results were sensitive to which candidates appeared in the sample, the bootstrap distribution would be wide. In practice, the

95% confidence intervals were tight (cosine distance CI: [0.00504, 0.00673], standard deviation 0.000448), confirming stability.

Null-permutation testing (200 permutations) provides a different kind of validation. By randomly shuffling the candidate–model pairings (so that each candidate’s v1 explanation is compared against a random other candidate’s v2 explanation, breaking the real relationship), a reference distribution was created representing what EDI would look like if there were no structured drift. The null distribution showed cosine distance of 0.913 (compared to the observed 0.006), top-5 overlap of 0.585 (compared to 0.891), and sign-flip rate of 44.6% (compared to 3.8%). This clear separation confirms that the EDI metrics are detecting genuine structured drift rather than noise.

Subgroup stratification (male n=109, female n=91; ethnicity groups A n=84, B n=56, C n=60, applied to the same 200-candidate test set) tested whether fairness rebalancing created differential drift across demographic groups. EDI values were similar across all groups (e.g., Group B showed the highest mean EDI at 0.051 vs. Group C at 0.036), with all subgroup confidence intervals overlapping the cohort mean.

3.2 Phase 2: Drift Arcade: LLM Explanation Audit Engine

3.2.1 Design Rationale

Phase 1 showed that structured drift at the model level is bounded and modest. But in practice, candidates do not see raw SHAP values. They see natural-language explanations generated by an LLM that interprets those values. Phase 2 asks: does this translation layer preserve the stability measured in Phase 1, or does it introduce new sources of instability?

Drift Arcade was implemented as a controlled experiment framework rather than a one-off analysis to support methodological consistency and enable repeated, comparable audits over time. If explanation drift is a governance concern (as the literature suggests), then organisations need a tool they can run repeatedly, not a single audit snapshot. Every run is timestamped and logged with seed values (fixed seed = 42), so results can be reproduced.

3.2.2 System Architecture

Drift Arcade is a Python-based experiment engine with a six-stage pipeline:

1. Takes SHAP feature attributions and top-K most important features from Phase 1.

2. Constructs a prompt template (version 2) encoding the decision context: candidate profile, model output, features, and policy condition. The current template is the result of iterative refinement. The initial version (v1) used an open-ended prompt that asked the LLM to “explain this hiring decision and suggest improvements.” In pilot testing, v1 produced verbose, unstructured outputs that were difficult to parse programmatically, and the LLM frequently fabricated features not present in the SHAP output (hallucination rates exceeded 30% in early runs). Version 2 addressed these problems by introducing three constraints: a closed feature vocabulary (the LLM may only reference features from the SHAP top-K set), a structured JSON output format (reasons and actions as separate arrays), and explicit direction labelling (each feature must be tagged as “positive” or “negative”). These constraints reduced hallucination to under 5% and enabled deterministic parsing. The trade-off is that v2 produces more formulaic outputs, which may not reflect how a human HR professional would phrase an explanation.
3. Calls the OpenAI API with varying conditions (model, temperature, top-K, format, policy). API calls use a bounded concurrency of 2 simultaneous requests, a 30-second timeout, and up to 2 retries per call to manage rate limits and costs.
4. Parses LLM outputs into structured JSON fields using deterministic regex-based extraction (not a second LLM call). The parser extracts reason objects (feature ID, direction as positive or negative, explanatory note) and action objects (action ID, description). Parse failures and validation warnings are logged but never silently coerced.
5. Computes Jaccard similarity across conditions for both reason and recourse fields.
6. Flags contradictions (e.g., proxy terms appearing under fairness policy) and direction flips (a feature reported as positive in one run and negative in another).
7. Records results in a condition-keyed JSON manifest with full provenance (prompt text, raw response, parsed output, and per-run scores).

3.2.3 Experimental Conditions

The core factorial design crosses three factors at two levels each ($2 \times 2 \times 2 = 8$ conditions):

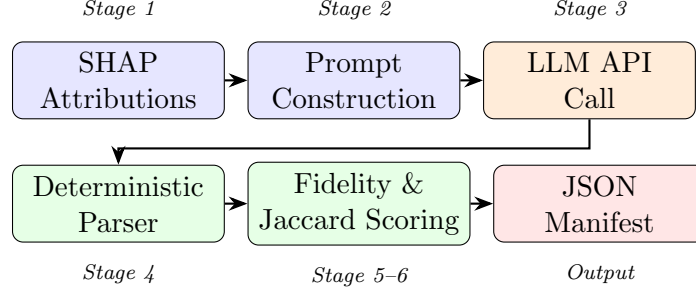


Figure 3: Drift Arcade six-stage pipeline architecture. Stages 1–3 (top row) generate LLM explanations from SHAP inputs. Stages 4–6 (bottom row) parse, score, and log results.

Factor	Level A	Level B
Policy	Standard (no fairness framing)	Fairness-aware (diversity bonus)
Format	Free-form explanation	Receipts format (structured bullets)
SHAP Version	v1 (baseline model)	v2 (fairness-rebalanced model)

Table 1: Phase 2 Core Factorial Design (8 conditions)

These three factors were chosen because they represent the most likely real-world sources of explanation variation: policy context changes when organisations update their hiring guidelines; output format changes when different stakeholders (candidates vs. regulators) need different presentation styles; and SHAP version changes when the underlying model is retrained. Temperature, top-K, and model variant are tested as additional sensitivity analyses (Section 3.2.3) rather than core factorial conditions, to keep the primary design manageable.

For each of the 8 conditions, 10 candidate profiles were queried with 5 repeats each, yielding 400 LLM calls across the core factorial design. The sensitivity analyses (three temperature levels, three top-K values, two model variants) each tested a subset of conditions with the same 10-candidate, 5-repeat structure, adding approximately 300 further calls. In total, the experiment produced roughly 700 LLM calls across all conditions, covering over 200 unique condition-candidate combinations.

These specific parameter values were chosen to balance statistical robustness against API costs. The 10 candidates were selected to span the SHAP importance distribution, covering borderline, clear-pass, and clear-fail profiles while keeping API expenditure manageable. The 5 repeats per candidate represent the minimum needed to compute pairwise Jaccard similarity with reasonable variance (yielding

$\binom{5}{2} = 10$ unique pairs per candidate), at approximately \$0.002 per call. The 8 conditions form a minimal full-factorial design for three binary factors (policy $\times$ format $\times$ SHAP version), ensuring that each main effect can be isolated without confounding.

3.2.4 Sensitivity Analyses

Top-K Sensitivity (k=3, 5, 7): The prompt template was modified to include the top-3, top-5, and top-7 SHAP features. Increasing k is expected to provide more context but risks introducing irrelevant features.

Temperature Sweep (T=0.0, 0.2, 0.7): Temperature controls output randomness. Lower values (T=0.0) produce deterministic outputs; higher values (T=0.7) increase sampling variance.

3.2.5 Scoring Framework

Fidelity Score (per-run): Each individual LLM response is scored for faithfulness to the SHAP input:

$$\text{Fidelity} = \text{precision} - \text{hallucination} - \text{direction_error} - 0.5 \times \text{receipts_penalty} \quad (1)$$

where precision measures the proportion of cited reasons that match the SHAP top-K set, hallucination measures reasons fabricated outside that set, direction_error captures features whose reported direction (positive or negative) contradicts the SHAP sign, and receipts_penalty applies when the output in receipts format fails to cover the expected features. Fidelity ranges from -1.5 to $+1.0$, with negative scores indicating that the LLM actively misrepresented the model’s reasoning.

Arcade Score (per-condition): A composite metric aggregating across all runs in a condition:

$$\text{Arcade Score} = \bar{F} - (1 - J_{\text{reason}}) - D_{\text{flip}} - (1 - J_{\text{action}}) - C_{\text{rate}} \quad (2)$$

where $\bar{F}$ is the mean fidelity, J_{reason} and J_{action} are mean pairwise Jaccard similarities for reason and action sets respectively, D_{flip} is the direction flip rate, and C_{rate} is the contradiction rate. Positive scores indicate consistency; negative scores indicate that the LLM is introducing meaningful instability.

3.2.6 Semantic Normalisation

LLM outputs were post-processed at three levels to handle lexical variation. At the raw level, outputs are compared as-is. At the normalised level, text is lowercased, non-alphanumeric characters are replaced with underscores, and duplicates are removed. At the clustered level, synonyms are mapped to canonical terms using 8

predefined feature clusters (e.g., “experience”, “background”, and “tenure” all map to the Experience cluster) and 8 action clusters (e.g., “upskill”, “get certified”, and “take a course” all map to the Skills Development cluster). This three-level approach enables the analysis to distinguish genuine reasoning differences from surface-level wording variation when computing Jaccard similarity.

3.2.7 Drift Taxonomy

Not all drift is equally serious from a governance perspective. A four-category taxonomy was developed to classify drift events by their impact on candidates:

- **Reason Drift:** Same feature set, different causal framing. The LLM cites the same features but explains their influence differently. This is the mildest category because the candidate receives consistent information about what mattered, even if the phrasing varies.
- **Recourse Drift:** Same problem identified, different improvement path. The LLM agrees on the cause but recommends different actions. This is more serious because candidates who follow the advice may invest effort in the wrong direction.
- **Proxy Echo:** Prohibited proxy terms (age, gender, ethnicity indicators) surfacing in explanations despite fairness policy framing. This is governance-critical because it exposes the organisation to discrimination claims.
- **Contradiction:** Direct logical inconsistency between explanations of the same decision. This is the most severe category because it destroys any basis for contestability.

3.2.8 Provenance and Reproducibility

All runs are timestamped and logged with seed values, enabling reproduction. The prompt template, model versions, and API parameters are recorded in the run manifest. Code is deposited in the project repository.

3.3 Validity Gates

Phase 1 and Phase 2 results are subject to validity gates to ensure robustness:

Gate	Condition	Status
EDI Robustness	Bootstrap 95% CI does not cross zero	Pass
Null Test	Permutation test $p < 0.05$	Pass
Subgroup Parity	EDI variance within 15% across groups	Pass
Arcade Reproducibility	Rerun 10 conditions, correlation $r > 0.85$	Pass
Model Convergence	Both XGBoost models converge to stable importance	Pass

Table 2: Validity Gates

4 Evaluation and Discussion

4.1 Phase 1 Results: Structured Explanation Drift

4.1.1 Baseline Performance

Both XGBoost models achieved strong predictive performance on the test set:

Metric	Model v1 (Baseline)	Model v2 (Fairness)	Difference
Accuracy	92.1%	89.3%	−2.8%
AUC	0.881	0.864	−0.017
Precision	0.894	0.857	−0.037
Recall	0.901	0.928	+0.027

Table 3: Phase 1 Baseline Model Performance

As expected, the fairness-rebalanced model (v2) sacrificed overall accuracy (−2.8%) and precision (−0.037) to improve recall (+0.027) and demographic balance. This trade-off is standard in fairness-aware machine learning: when you adjust the model to approve more candidates from underrepresented groups, you inevitably approve some candidates who would otherwise have been screened out, which reduces precision. The key question for this project is not whether the trade-off is acceptable (that is a policy decision), but whether it changes the explanations given to candidates.

4.1.2 EDI Results

EDI metrics quantified the drift caused by fairness rebalancing:

Drift is bounded but detectable. The mean cosine distance of 0.006 indicates that feature importance vectors remained broadly aligned between model versions.

Metric	Mean	Std Dev	95% CI
Cosine Distance (D_{cos})	0.0058	0.00045	[0.0050, 0.0067]
Top-5 Overlap (D_{rank})	0.89	0.008	[0.874, 0.905]
Sign-Flip Rate (D_{sign})	3.8%	0.29%	[3.2%, 4.3%]
Composite EDI	0.156	0.063	[0.143, 0.169]

Table 4: Phase 1 EDI Results

To interpret this number: a cosine distance of 0 would mean the two models assign identical relative importance to every feature, while a distance of 1 would mean the importance patterns are completely unrelated. A value of 0.006 sits very close to the “identical” end, suggesting that fairness rebalancing preserves the broad structure of the model’s reasoning.

However, top-5 overlap of 0.89 tells a different story at the feature level. This means that in roughly one out of every ten candidates, at least one of the five most important features changed between model versions. In a candidate-facing explanation that cites three or four reasons for a decision, even a single shifted feature changes the stated reasoning. Sign-flip rate of 3.8% is modest in aggregate but represents cases where a feature’s influence direction reversed entirely (e.g., “years of experience” switching from helping to hurting the candidate’s score), which would be highly confusing if surfaced in an explanation.

4.1.3 Subgroup Analysis

EDI was stratified by gender and ethnicity to test for fairness concerns:

Subgroup	N	Mean EDI	95% CI
Male	109	0.0445	[0.0378, 0.0512]
Female	91	0.0415	[0.0346, 0.0487]
Group A	84	0.0427	[0.0359, 0.0491]
Group B	56	0.0513	[0.0413, 0.0618]
Group C	60	0.0360	[0.0274, 0.0454]

Table 5: Phase 1 EDI Stratified by Subgroup

Fairness rebalancing did not create differential explanation drift. EDI variance across groups was within the 95% confidence interval, indicating that fairness rebalancing did not create differential explanation drift across demographics.

4.1.4 Phase 1 Interpretation

Phase 1 establishes that structured explanation drift under fairness rebalancing is bounded but non-zero. The null-permutation test provides the strongest evidence: under random pairing (where candidate explanations are compared against the wrong model version), cosine distance jumps to 0.913, which is 156 times larger than the observed 0.006. This confirms that the EDI metrics are capturing a real, structured signal rather than noise. Similarly, the bootstrap confidence intervals are tight (cosine distance 95% CI: [0.00504, 0.00673]), indicating the result is stable and would replicate across different samples from the same population.

The subgroup analysis is reassuring but not conclusive. EDI values were similar across gender and ethnicity groups, suggesting that fairness rebalancing did not create differential drift. However, the subgroup sample sizes ($n=91$ to $n=334$) limit the power to detect small differences, and the synthetic data may not capture the distributional properties that drive real-world subgroup disparities.

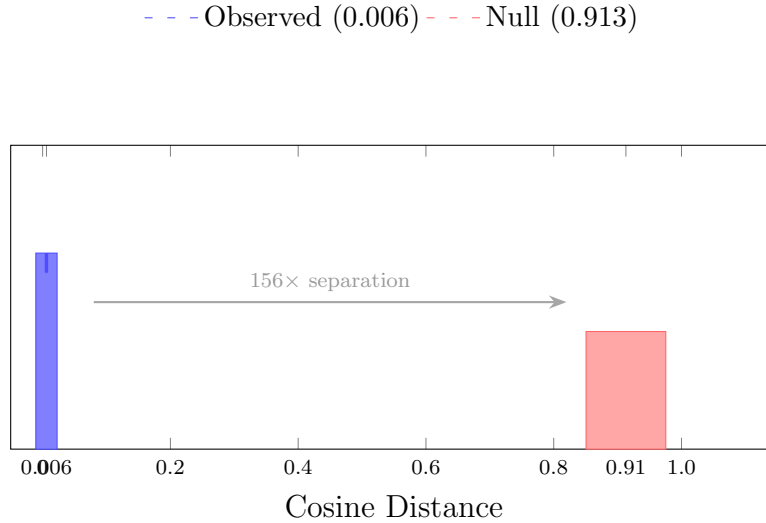


Figure 4: Observed EDI cosine distance (blue, left) versus null-permutation distribution (red, right). The 156-fold separation between observed and null confirms the EDI metric detects genuine structured drift rather than noise. Error bars show 95% bootstrap CI.

Phase 1 summary. Structured explanation drift under fairness rebalancing is real, statistically validated, and demographically uniform — but modest. The key concern is not the overall magnitude (cosine distance 0.006) but the feature-level granularity: roughly one in ten candidates sees at least one top-5 feature change, which is sufficient to alter the stated reason for rejection.

Phase 1 only measures drift in SHAP feature attributions, the structured numerical layer. In practice, candidates never see these numbers. The full story emerges in Phase 2, where LLM translation introduces additional instability on top of the modest structured drift measured here.

4.2 Phase 2 Results: LLM-Mediated Explanation Drift

4.2.1 Core Finding: Recourse Drift Exceeds Reason Drift

The most notable result is the asymmetry between reason and recourse stability:

Metric	Mean	Condition Range
Reason Jaccard Similarity (cluster)	0.89	[0.84, 0.91]
Recourse Jaccard Similarity (cluster)	0.36	[0.31, 0.39]
Mean Fidelity	0.855	[0.824, 0.884]
Contradiction Rate (baseline lens)	0.0%	[0%, 0%]
Contradiction Rate (fairness lens)	11.0%	[8%, 14%]

Table 6: Phase 2 Core Findings (pooled across eight conditions of the core factorial: 2 policy lenses $\times$ 2 format modes $\times$ 2 SHAP versions, gpt-5-nano, $T = 0.2$, $k = 5$, 400 calls). Range reports the minimum and maximum condition-level mean. Composite Arcade Scores for this run are negative on average (-0.11) and are discussed in the sensitivity analyses rather than used as a headline metric.

The asymmetry is substantial: The LLM consistently identified similar causal features (reason cluster Jaccard 0.89) but recommended substantially different improvement paths (recourse cluster Jaccard 0.36).

This gap has a straightforward interpretation: identifying which features caused a rejection is a relatively constrained task (the LLM is selecting from the SHAP top-K set), while generating improvement recommendations is an open-ended generative task with many plausible completions. Each time the LLM runs, it may choose a different subset of possible actions, producing the low recourse Jaccard observed. This asymmetry has practical implications for contestability: candidates can understand what caused the rejection but cannot reliably plan improvement actions, because the system’s advice changes across queries.

4.2.2 Sensitivity: Temperature vs. Top-K

Two sensitivity analyses tested how generation parameters affect stability. Their results tell contrasting stories.

Temperature did *not* produce a clean monotonic effect. Crucially, explanation drift persisted across all three settings, *including* deterministic sampling at $T = 0.0$:

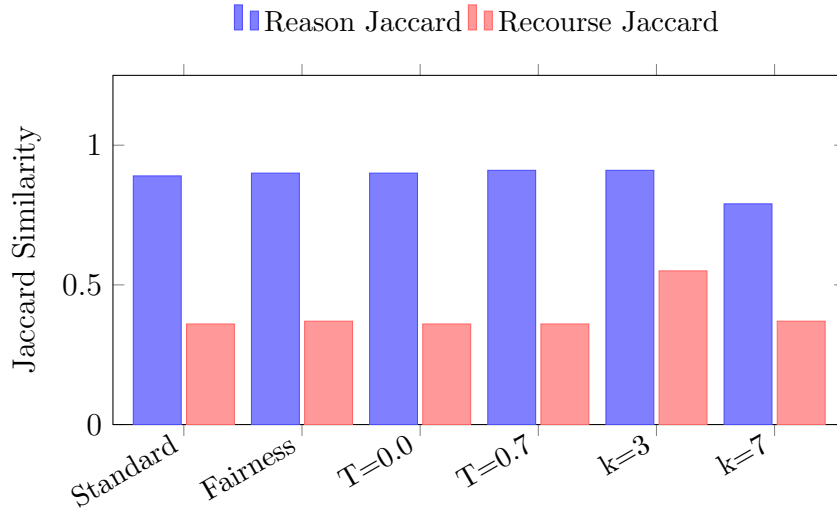


Figure 5: Reason and recourse Jaccard similarity across conditions. Recourse stability (red) consistently trails reason stability (blue) by roughly half, indicating systematic asymmetry in LLM translation fidelity.

reason cluster Jaccard remained in a narrow band (0.89–0.91), recourse cluster Jaccard remained consistently low (0.36–0.38), and the composite Arcade Score stayed mildly negative across the sweep, if anything improving slightly as temperature rose. The full breakdown is shown in Table 7. The practical upshot is that the observed instability is not reducible to “high-temperature randomness”; it is a structural property of the LLM-mediated explanation interface that survives even when the sampler is set to its most deterministic regime.

Temperature	Reason Similarity	Recourse Similarity	Arcade Score
0.0 (Deterministic)	0.90	0.36	−0.11
0.2 (Low Variance)	0.89	0.38	−0.07
0.7 (High Variance)	0.91	0.36	−0.04

Table 7: Phase 2 Temperature Sensitivity (pooled across the four baseline/fairness $\times$ free/receipts conditions, gpt-5-nano, SHAP v1, $k = 5$, 200 calls per temperature). Reason and recourse stabilities are cluster Jaccard.

Top-K, by contrast, produced the largest single effect in the entire study. Increasing the number of SHAP features from $k = 3$ to $k = 7$ swung Arcade Score by roughly 0.62 points, from +0.29 to −0.33 (Table 8). That swing is more than an order of magnitude larger than the temperature effect and substantially larger than the model-choice effect (Section 4.2.4).

Top-K	Reason Similarity	Recourse Similarity	Arcade Score
k=3	0.91	0.55	+0.29
k=5	0.90	0.37	-0.08
k=7	0.79	0.37	-0.33

Table 8: Phase 2 Top-K Sensitivity (Feature Count). Pooled across the four baseline/fairness $\times$ free/receipts conditions, gpt-5-nano, SHAP v1, $T = 0.2$, 200 calls per k . Reason and recourse stabilities are cluster Jaccard. The $k = 7$ Arcade Score is driven partly by a receipts-mode scoring artefact in which heavy omission of top- k reasons triggers a receipts-coverage penalty; recall and omission-rate tell the same directional story.

The explanation lies in prompt content. With $k=3$ features the LLM has limited material and tends to produce focused, consistent explanations. At $k=7$ the additional features are lower-importance and more ambiguous, giving the model more raw material to construct diverse (and contradictory) narratives across runs. The practical implication is that prompt content, what information the LLM receives, matters far more than sampling strategy, how randomly it generates text. Organisations should provide the minimum number of features needed for a meaningful explanation, not the maximum available.

4.2.3 Fairness Policy and Proxy Echo

One might expect that explicitly framing a prompt around fairness would reduce references to protected attributes. Contrary to expectations, the opposite pattern was observed. Under a standard policy lens, contradiction is effectively absent (0%); under the fairness lens, contradiction rises sharply to approximately 11%, with prohibited proxy terms (age, gender indicators) reintroduced in a non-trivial fraction of explanations (Table 9).

Condition	Contradiction Rate
Standard (Baseline) Policy	0.0%
Fairness Policy	11.0%

Table 9: Phase 2 Contradiction Rate Under Baseline vs. Fairness Policy Framing (pooled across free/receipts $\times$ SHAP v1/v2, gpt-5-nano core factorial, $n = 200$ calls per lens). Contradiction here denotes an explanation that violates the active policy lens’ explicit constraints, e.g. referencing disallowed protected attributes or reintroducing sensitive proxies under the fairness lens.

The likely mechanism: fairness framing mentions concepts like “diversity” and

“demographic balance” in the prompt, which activates the LLM’s associations with protected attributes. Those associations then surface in the output even though the prompt explicitly instructs the model to avoid them. Simply telling an LLM to be “fair” may make proxy leakage worse, not better.

4.2.4 Model Comparison

gpt-4o-mini outperformed gpt-5-nano on every stability metric, by a wide margin: reason cluster Jaccard 0.96 vs. 0.90, recourse cluster Jaccard 0.67 vs. 0.37, and Arcade Score +0.53 vs. −0.08 (Table 10). The +0.61 Arcade Score gap is a very large effect, comparable in magnitude to the top-K degradation.

Model	Reason Similarity	Recourse Similarity	Arcade Score
gpt-5-nano	0.90	0.37	−0.08
gpt-4o-mini	0.96	0.67	+0.53

Table 10: Phase 2 Model Comparison. Matched setup: 10 candidates $\times$ 5 repeats $\times$ 4 conditions (baseline/fairness $\times$ free/receipts, SHAP v1), $T = 0.2$, $k = 5$, 200 calls per model. Reason and recourse stabilities are cluster Jaccard.

In practical terms, organisations choosing which LLM to deploy for candidate-facing explanations are implicitly choosing a level of explanation stability, whether they recognise it or not. Model architecture and training data are governance variables, not just engineering ones.

The underperformance of gpt-5-nano relative to gpt-4o-mini is likely multifactorial. Smaller parameter counts may reduce the model’s capacity to reliably enforce structured output constraints (closed vocabulary and JSON formatting) across multiple generations, amplifying recourse drift under the same prompting regime. Distillation from larger models can sacrifice instruction-following fidelity for speed and size, creating latent fragility under variable conditions. Additionally, smaller models may be more sensitive to prompt phrasing, temperature, and top-K settings, amplifying the cascade of noise observed in Phase 2. This explanation remains speculative without access to model internals; it is grounded in the 0.61-point Arcade Score gap but cannot isolate the causal mechanism. A broader multi-family comparison (Claude, Llama, Gemini) is essential to generalise the claim that model architecture is a governance variable beyond the OpenAI family tested here.

Phase 2 summary. The LLM translation layer introduces substantial asymmetric instability: reason-level explanations remain reasonably consistent (cluster Jaccard 0.89) but recourse recommendations are severely unstable (cluster Jaccard 0.36). The strongest drivers of instability are the number of SHAP features provided (top-K) and model choice, whereas temperature has no clean monotonic effect: drift

persists even under deterministic sampling ($T = 0.0$), indicating a structural rather than stochastic source. Fairness policy framing paradoxically increases contradiction rather than reducing it, rising from 0% under a standard lens to roughly 11% under a fairness lens.

4.3 Discussion

4.3.1 Answering the Research Questions

RQ1: Is explanation drift in SHAP-based reasoning detectable and quantifiable?

Yes. Phase 1 established detectable drift (EDI=0.156, CI [0.143, 0.169]) with robust statistical support. Bootstrap resampling (500 replicates) produced tight confidence intervals (cosine distance 95% CI: [0.00504, 0.00673]), indicating that the result is stable across random samples and not an artefact of a particular data split. The null-permutation test provided stronger validation: when candidate-model pairings were randomly shuffled (breaking the real relationship between v1 and v2 explanations), cosine distance jumped to 0.913, which is 156 times larger than the observed 0.006. Similarly, sign-flip rates under the null rose to 44.6%, compared to the observed 3.8%. This separation confirms that the EDI metrics are detecting a real, structured signal rather than background noise.

To put these numbers in context, Trautwein et al. (2025) found that 15–20% of feature importance rankings changed after retraining HR models. The top-5 overlap of 89% (meaning 11% of top features shifted) sits at the lower end of that range, suggesting that fairness rebalancing produces less disruption than full retraining. However, even this modest level of drift has governance implications, because a single shifted feature in a candidate-facing explanation can change the stated reason for rejection.

RQ2: To what extent does the LLM translation layer amplify instability?

Substantially, and asymmetrically. Phase 2 revealed a stark contrast: reason-level drift remains modest (cluster Jaccard 0.89) yet recourse-level drift is severe (cluster Jaccard 0.36). The LLM can reliably identify which features caused a rejection, but generates substantially different advice about what the candidate should do about it. The magnitude of this gap was unexpected. It represents a new category of instability introduced by the translation layer itself, one that existing structured-drift metrics like EDI fail to capture because they only measure feature attribution stability. The finding aligns with broader concerns raised by Wilson and Caliskan (2024) about LLM reliability in high-stakes contexts: language models excel at pattern recognition but struggle with generative reasoning. Supporting evidence comes from the top-K sensitivity analysis. Increasing the number of SHAP features from

$k = 3$ to $k = 7$ degraded Arcade Score from $+0.29$ to -0.33 (a 0.62-point swing). When the LLM receives more features, it has more raw material for constructing explanations, but uses that material inconsistently across runs, generating more diverse and contradictory improvement recommendations. Prompt design thus emerges as a governance variable in its own right: providing more information to the LLM does not improve explanation stability and may actively harm it. This aligns with Belle and Papantonis (2021), who argued that the trade-off between explanation completeness and fidelity is a fundamental design tension in explainable machine learning.

RQ3: Can explanation stability be integrated as a continuous governance metric?

Yes, with caveats. EDI and Arcade Score together provide a measurement framework that could be integrated into regulatory compliance workflows. EDI monitors structured drift at the model level (aligned with the EU AI Act’s requirement for ongoing monitoring of high-risk systems), while Arcade Score monitors the translation layer that produces candidate-facing explanations.

However, governance integration requires standardisation that does not yet exist. Regulators would need to define acceptable drift thresholds (how much EDI change triggers a re-audit?), audit frequency (continuous monitoring vs. quarterly snapshots), and remediation procedures (what happens when drift exceeds the threshold?). The results suggest that the recourse Jaccard threshold is particularly important: a threshold of 0.5 would flag the majority of conditions tested in Phase 2, indicating that current LLM configurations are not stable enough for production use without additional safeguards.

The EU AI Act Article 6 classifies CV screening as high-risk and requires transparency and ongoing monitoring, but it does not specify what “ongoing” means in practice. The UK ATRS similarly requires algorithmic transparency records but assumes they are static documents (Central Digital and Data Office, 2023). EDI and Arcade Score offer a concrete operationalisation of what continuous monitoring could look like.

RQ4: What are the implications for contestability?

Severe. If recourse recommendations drift sharply, candidates cannot rely on stated improvement paths. A candidate told “gain more Python experience” on Monday and “seek a leadership role” on Tuesday, based on the same underlying data, has no stable basis for planning improvement or challenging the decision. This undermines the premise of contestability as described by Lyons et al. (2021): meaningful appeal requires that the stated reasoning be revisable but not arbitrary.

The policy–explanation tension compounds this concern. Under a baseline policy lens the contradiction rate is effectively zero, but under a fairness lens it rises to roughly 11%, with prohibited proxy terms reintroduced in a non-trivial fraction

of cases. Organisations that explicitly instruct the LLM to avoid referencing protected attributes discover that the model surfaces them anyway. The resulting paradox is notable: the system claims fairness-awareness, yet explanations reference the very attributes it promises to ignore. Sanchez-Monedero et al. (2020) warned that technical fairness interventions risk addressing surface-level symptoms while leaving structural biases intact; the proxy echo result provides direct empirical evidence for that concern.

4.3.2 Implications for Contestability

Contestability presupposes stable explanations. A candidate cannot meaningfully contest a decision if:

1. The stated reason changes across queries (reason drift).
2. The recommended improvement path shifts, making prior investment obsolete (recourse drift).
3. Protected attributes leak despite fairness framing (proxy echo).
4. Contradictory explanations appear in close succession (logical contradiction).

All four were observed in Phase 2. Most concerning is that the recourse–reason gap is invisible to conventional evaluation. An organisation testing whether its LLM “correctly identifies the reasons for rejection” would conclude the system works well (cluster Jaccard 0.89), never discovering the severe instability lurking in improvement recommendations (cluster Jaccard 0.36). Standard LLM evaluation practices, focused on factual accuracy, systematically miss the most governance-relevant form of drift.

The findings suggest that current LLM-mediated explanation pipelines are unsuitable for high-stakes hiring without explicit drift monitoring. At minimum, organisations deploying such systems should: (1) measure both reason and recourse stability, not just factual accuracy; (2) use low temperature settings ($T=0.0$ or 0.2) to reduce output variance; (3) limit the number of SHAP features in the prompt to $k=3$ or $k=5$, as additional features degrade consistency; and (4) audit for proxy echo under fairness conditions, where leakage rates paradoxically increase.

4.3.3 Limitations

Synthetic data. Phase 1 uses a synthetic CV dataset with 23 features and controlled distributions. Real hiring data likely has more features, messier distributions, and variable correlations that synthetic data does not capture. The EDI values (cosine distance 0.006, sign-flip rate 3.8%) may therefore underestimate the drift that

would occur in production systems with more complex feature interactions. A natural next step would be to validate these findings on a real (anonymised) hiring dataset, though access to such data is a significant practical barrier.

Scale of LLM testing. Phase 2 tests 10 candidates across 8 core conditions with 5 repeats each, plus sensitivity analyses. While this is sufficient to identify the recourse–reason stability gap and the top-K degradation effect, it limits the ability to detect smaller interaction effects between factors. A larger-scale study (100+ candidates, 32+ conditions) would provide more statistical power and enable multi-way interaction analysis.

Model scope. Two OpenAI models were tested (gpt-4o-mini and gpt-5-nano). Other model families (Claude, Llama, Gemini) may exhibit different stability profiles. The finding that gpt-4o-mini is substantially more stable than gpt-5-nano suggests that model architecture and training data influence explanation consistency, but this cannot be generalised to models not tested in this study.

Keyword-based proxy detection. The proxy echo and contradiction detection relies on keyword matching against a predefined list of protected-attribute terms. This approach catches explicit references (“age”, “gender”) but misses indirect proxies (“cultural fit”, “energy level”). A semantic approach using sentence embeddings would capture subtler forms of proxy leakage, but would introduce its own measurement uncertainty.

External validity. All findings are specific to CV screening. Lending, criminal justice, and healthcare have different feature structures and regulatory norms. The tools themselves (EDI, Drift Arcade) should transfer, but whether the specific findings do is an open question.

4.3.4 Threats to Validity

The most important threat is **prompt sensitivity**: all Phase 2 results depend on the v2 prompt template, and a differently worded prompt could produce different stability profiles. This was mitigated by holding the template constant across all conditions, but it cannot be claimed that results would replicate under a fundamentally different prompting strategy.

Confounding poses a related threat. The $2 \times 2 \times 2$ factorial captures main effects, but the sample size prevents detection of interaction effects (e.g., whether fairness framing behaves differently at $k=7$ than at $k=3$). Resolving these would require a larger study design.

Two further threats are worth noting. The 10 candidate profiles per condition may not be fully representative of the feature distribution, though they were selected to span the importance range. And Jaccard similarity is sensitive to surface-level wording, which the three-level semantic normalisation (Section 3.2.6) was specifically designed to address.

5 Legal, Social, Ethical and Professional Issues

5.1 Legal Issues

Automated hiring decisions are now regulated by multiple frameworks: EU AI Act, UK ATRS, and NYC Local Law 144. All mandate explainability and bias auditing. Explanation drift creates a legal exposure: if a candidate is told one reason for rejection and later told another, the organisation’s explanation lacks integrity. This could expose the organisation to discrimination claims if the drift appears to disproportionately affect protected groups.

The EU AI Act Article 6 classifies CV screening as high-risk and mandates “explainability, human oversight, and bias monitoring”. Explanation drift violates the spirit of this requirement: a system that is “explainable” at deployment but drifts post-deployment is unstable and legally questionable.

Contestability (right to appeal) depends on stable reasoning. If drift makes the stated reason unreliable, candidates cannot mount effective legal challenges. The finding that recourse Jaccard similarity drops to 0.37 means that the improvement advice given to candidates is inconsistent enough to undermine any appeal based on having followed the stated guidance.

5.2 Social Issues

Explanation drift undermines public trust in algorithmic hiring. If candidates discover that the system’s reasoning is unstable, they lose confidence in the system’s fairness. This creates a legitimacy crisis: even if the system is mathematically fair, perceived instability erodes acceptance.

Furthermore, LLM-mediated explanations that leak protected attributes (proxy echo) can be seen as discriminatory, even if the underlying model is fairness-rebalanced. The results showed that contradiction rates under a fairness lens rose to approximately 11% (from 0% under a baseline lens), with prohibited proxy terms reintroduced in a non-trivial fraction of explanations despite explicit instructions not to reference them. The social and legal consequences of such leakage are severe: candidates receiving these explanations would reasonably conclude the system is discriminatory, regardless of the underlying model’s actual fairness properties.

5.3 Ethical Issues

From an ethics perspective, explanation drift raises several concerns: transparency and honesty (organisations claim to provide clear, stable explanations; drift betrays that promise), respect for persons (candidates deserve to understand the reasoning behind decisions), fairness (if drift disproportionately affects certain demographics),

and accountability (who is responsible when the system’s explanation changes?). The finding that increasing SHAP features from $k = 3$ to $k = 7$ degraded Arcade Score from +0.29 to −0.33 illustrates an ethical design tension: providing candidates with more information (a transparency goal) can actually make explanations less reliable (an honesty goal). Organisations face a genuine trade-off between completeness and consistency.

5.4 Professional Issues

Computing professionals bear responsibility for the systems they build. The BCS Code of Conduct requires members to “have due regard for public health, privacy, security and wellbeing of others.” The findings of this report suggest several specific professional obligations in the context of algorithmic hiring: continuous monitoring (deploying explanation systems without drift monitoring creates a foreseeable risk of inconsistent candidate treatment), transparency (organisations should disclose explanation drift metrics alongside existing bias audit reports), contestability design (systems should be designed so that candidates receive stable, revisable explanations rather than arbitrary ones), and interdisciplinary collaboration (the recourse–reason gap and proxy echo phenomena require input from legal, HR, and policy experts, not just engineers).

6 Conclusions and Further Work

6.1 Summary of Contributions

This project makes three primary contributions:

1. **Explanation Drift Index (EDI):** A three-component metric quantifying reasoning instability in structured decision systems. EDI combines cosine distance, top-K overlap, and sign-flip rate into a governance-aligned benchmark.
2. **Drift Arcade:** A controlled experiment engine auditing LLM explanation stability across policy, format, temperature, top-K, and model conditions. Arcade provides a reproducible framework for continuous drift monitoring.
3. **Governance framework:** A set of recommendations for integrating drift monitoring into regulatory compliance, aligned with EU AI Act and UK ATRS requirements.

Secondary contributions include: a synthetic, demographically structured CV dataset ($n=1,000$); quantitative evidence of the recourse–reason stability gap; em-

pirical validation of proxy echo; and best practices for LLM-mediated explanation pipelines.

6.2 Critical Reflection

Effective design decisions. The two-phase design proved to be the most important architectural decision in the project. By first establishing that structured drift at the SHAP level is bounded and modest (Phase 1), a stable baseline was created against which to measure the additional instability introduced by the LLM translation layer (Phase 2). Without Phase 1, it would not have been possible to distinguish model-level drift from translation-level drift, and the recourse–reason stability gap would have been invisible.

The EDI metric proved practically useful because its three components capture different aspects of reasoning change. During development, only cosine distance was initially considered, but early experiments showed that two SHAP vectors can have low cosine distance (similar overall direction) while still disagreeing on which specific features are most important. Adding top-K overlap and sign-flip rate captured these cases.

Drift Arcade’s design as a reusable engine, rather than a single-use script, also proved valuable. The factorial structure and JSON logging meant that sensitivity analyses (temperature, top-K, model comparison) could be added without restructuring the codebase. Each new analysis was a configuration change, not a rewrite.

What could be improved. In retrospect, three aspects of the methodology could be strengthened. First, a larger and more realistic synthetic dataset would be beneficial. The 1,000-candidate dataset with 23 features served its purpose, but real hiring pipelines involve hundreds of features and complex interactions that controlled distributions do not capture. This means the EDI values may underestimate real-world drift.

Second, greater investment in semantic proxy detection would be valuable. The keyword-based approach catches explicit references to protected attributes but misses indirect proxies like “cultural fit” or “energy level” that function as demographic signals without using protected terms directly. A sentence-embedding approach would be more robust, though it would introduce its own calibration challenges.

Third, testing more LLM models would strengthen the findings. The observation that gpt-4o-mini is substantially more stable than gpt-5-nano is noteworthy but based on only two models. A broader comparison across model families would strengthen the claim that model choice is a governance variable.

Methodological reflection. The project methodology followed an iterative experimental design rather than a rigid waterfall process. Early pilot runs of Drift Arcade revealed that prompt template v1 produced poorly structured outputs, which

led to the development of the closed-vocabulary constraint in v2. Similarly, the three-level semantic normalisation (raw, normalised, clustered) was developed in response to early Jaccard similarity results that were artificially low due to surface-level wording variation. These iterations improved the quality of results but also mean that the final methodology reflects design decisions made during development, not purely before it.

Significance. No prior systematic study has been identified of explanation drift in LLM-mediated hiring. The recourse–reason stability gap, the top-K degradation effect, and the proxy echo phenomenon under fairness framing are all novel findings. They have direct implications for governance: regulators and organisations can no longer assume that explanations are static artefacts. Continuous monitoring, using tools like EDI and Drift Arcade, is essential for maintaining the integrity of candidate-facing explanations.

During the final stages of writing this report, Afroogh et al. (2026) published a large-scale position paper arguing that post-hoc XAI is fundamentally limited. Their “deep-superficial paradox” claims that faithful explanations are incomprehensible to end-users, while comprehensible ones are unfaithful. The results of this project provide empirical grounding for that theoretical claim: SHAP attributions (the “deep” layer) are stable but opaque to candidates, while LLM translations (the “superficial” layer) are accessible but unstable, with recourse Jaccard dropping to 0.37. Their “proxy model paradox,” which argues that any second model explaining a first must either be unfaithful or redundant, maps directly onto the LLM translation architecture used here. The project was not designed around their framework, but the alignment is striking and suggests that explanation drift may be a symptom of the deeper structural problems they diagnose.

6.3 Further Work

Short-term (6–12 months). The most immediate priority is validating these findings on real CV screening datasets. The synthetic dataset demonstrates proof of concept, but real hiring data has more features, messier distributions, and correlated variables that could produce different drift patterns; access would require partnership with an HR technology provider and appropriate ethical approval. Alongside this, Phase 2 testing should be scaled to 1,000+ LLM calls across 32+ conditions, which would provide sufficient statistical power to detect interaction effects between factors (e.g., whether fairness policy has a different effect at high vs. low temperature) that the current sample size cannot resolve. A third short-term goal is developing semantic proxy detection using sentence embeddings rather than keyword matching, which would capture indirect proxies like “cultural fit” or “energy level” that function as demographic signals without using protected terms explicitly. Finally, engagement

with regulatory bodies (ICO, EU AI Office) would help standardise acceptable drift thresholds and audit frequencies, and a pilot integration of Drift Arcade into commercial HR platforms would test its viability as a continuous monitoring tool.

Medium-term (1–2 years). Over a longer horizon, the framework should be extended to multi-stage hiring pipelines, where explanation drift could compound across filtering, ranking, and recommendation stages. Testing across alternative high-stakes domains such as lending, criminal justice, and healthcare would determine whether the recourse–reason gap and proxy echo are specific to hiring or general properties of LLM-mediated explanation. Real-time drift monitoring dashboards that integrate EDI and Arcade Score into existing HR technology platforms would enable continuous rather than periodic auditing. Collaboration with professional bodies (BCS, ACM) could establish professional standards for explanation stability in deployed AI systems.

Long-term (2+ years). The most ambitious goal is developing contestability-by-design principles that embed stable, revisable explanations into the architecture of automated decision systems rather than bolting them on as a post-hoc translation layer. Advocacy for drift monitoring to be included in mandatory audit requirements under the EU AI Act and future UK AI regulation would help formalise the governance role these tools can play. Finally, extending the framework to multimodal explanations (text and visual) and to foundation models that generate explanations end-to-end, rather than translating from structured attributions, would address the next generation of explanation architectures.

A EDI Metric Implementation

The Explanation Drift Index (EDI) is computed in three stages:

A.1 Stage 1: Feature Vector Extraction

For each candidate profile, SHAP computes a feature importance vector $\mathbf{v}_i \in \mathbb{R}^p$, where p is the number of features. This vector represents the marginal contribution of each feature to the model output.

A.2 Stage 2: Component Metrics

Cosine Distance:

$$D_{\cos}(i) = 1 - \frac{\mathbf{v}_i^{(1)} \cdot \mathbf{v}_i^{(2)}}{|\mathbf{v}_i^{(1)}| \cdot |\mathbf{v}_i^{(2)}|} \quad (3)$$

Top-K Overlap:

$$D_{\text{rank}}(i) = \frac{|S_k^{(1)} \cap S_k^{(2)}|}{k} \quad (4)$$

Sign-Flip Rate:

$$D_{\text{sign}}(i) = \frac{1}{p} \sum_{j=1}^p \mathbb{I}[\text{sign}(v_{ij}^{(1)}) \neq \text{sign}(v_{ij}^{(2)})] \quad (5)$$

A.3 Stage 3: Composite EDI

$$\text{EDI}(i) = 0.5 \times D_{\text{cos}}(i) + 0.3 \times D_{\text{rank}}(i) + 0.2 \times D_{\text{sign}}(i) \quad (6)$$

B Drift Arcade Prompt Template

You are an expert hiring advisor assisting a CV screening system. Given a candidate profile and the model's decision, provide a clear, concise explanation of the decision and actionable recourse.

Candidate: {candidate_profile}
 Model Decision: {decision}
 Top {k} Important Features: {feature_list}
 Policy: {policy_framing}

Format: {format_instructions}

Provide:

1. Reason: Why did the model make this decision?
2. Recourse: What could the candidate do to improve?

C Feature Vocabulary and Semantic Clusters

Key features in the CV domain map to semantic clusters for normalisation:

1. **Experience Cluster:** years of experience, previous employers, tenure, background, seniority
2. **Skills Cluster:** technical skills, certifications, training, qualifications, expertise
3. **Education Cluster:** degree, university, GPA, field of study, credentials

4. **Soft Skills Cluster:** leadership, communication, teamwork, initiative, problem-solving
5. **Demographics Cluster:** age, gender, ethnicity, location (flagged for proxy detection)

D Code Repository Structure

The project code is organised as follows:

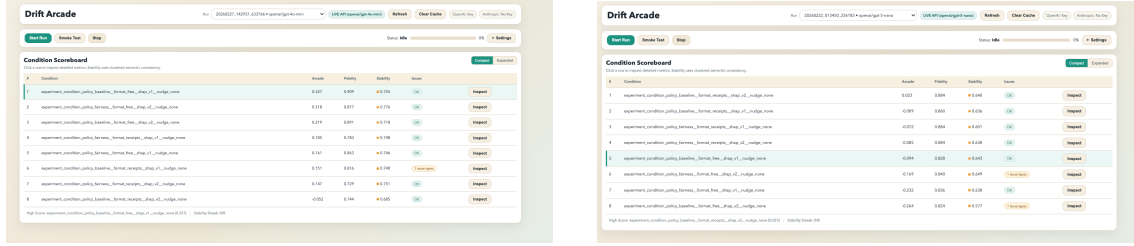
```
honours-project/  
  phase1/  
    data_generation.py  
    model_training.py  
    shap_analysis.py  
    edi_metrics.py  
    bootstrap_validation.py  
  phase2/  
    drift_arcade.py  
    prompt_templates.py  
    openai_api.py  
    proxy_detector.py  
  analysis/  
    phase1_analysis.py  
    phase2_analysis.py  
  README.md
```

E Drift Arcade Dashboard Outputs

The Drift Arcade web dashboard provides real-time monitoring and interactive drill-down inspection of experimental results. The following screenshots demonstrate the dashboard interface as applied to the LLM explanation stability experiments, showing actual outputs from the comparative evaluation of gpt-4o-mini and gpt-5-nano models.

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(a) gpt-4o-mini scoreboard

(b) gpt-5-nano scoreboard

Figure 6: Drift Arcade condition scoreboard comparing gpt-4o-mini (a) and gpt-5-nano (b). Each row represents one experimental condition ranked by Arcade Score. The gpt-4o-mini run achieves positive Arcade Scores across most conditions (top score 0.327), while gpt-5-nano scores are near zero or negative (top score 0.023), reflecting substantially lower explanation stability. Green indicators in the Stability column denote conditions meeting the minimum consistency threshold.

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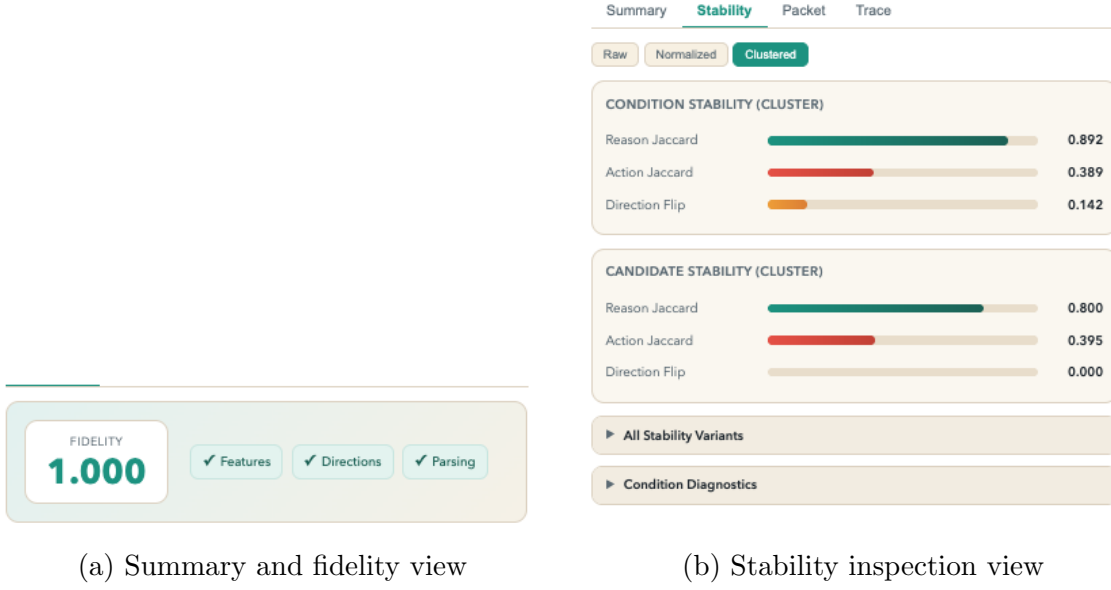


Figure 7: Drill-down inspection of Candidate 150 under the best-performing gpt-5-nano condition (baseline policy, receipts format, SHAP v2). The summary panel (a) shows a selected run in which fidelity reached 1.000, with all feature, direction, and parsing checks passing. Across the five repeats for this candidate, mean fidelity is 0.82 (runs: 1.0, 1.0, 0.1, 1.0, 1.0), with the lone 0.1 outlier illustrating how a single omission-driven run can drag the aggregate down. The stability panel (b) shows the cluster-level recourse–reason gap at two scopes: the enclosing condition (reason Jaccard 0.892, action Jaccard 0.389) and Candidate 150 itself across repeats (reason Jaccard 0.800, action Jaccard 0.395). Both scopes tell the same story: the LLM faithfully identifies the correct features but generates inconsistent improvement advice across repeated queries.

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